✨ **Prepared By** ✨

Zahidul Islam Joty

🎓 *Dept. (Textile Engineering Management)*
Batch — 48
⇨ AM-II (Dry wash process)

Sand Blasting –

What is Sand Blasting?
Sand blasting is a **dry process** where **high-speed air** mixed with **fine sand particles** is blown on jeans or garments. This is a very successful and most widely used process for fashion articles. When the surface area of the garment is blasted,This removes the top color from the fabric and shows the **white cotton inside**.

☞ It makes the jeans look **old, used, and stylish** (like naturally worn jeans).

Benefits of Sand Blasting:

✓ Makes a **smooth and even faded effect**.

✓ Gives a **natural worn look** that **looks real**.

✓ Very **popular** and **commonly used** in the fashion industry.

✓**Mechanical process**

✓Uses **Aluminum Oxide (Al_2O_3)**, which looks like **sand**.

✓Sand blasting is usually **combined with other washes**, like:

- Normal Wash
- Stone-Enzyme Wash
- Enzyme Wash
- Bleach Wash

<u>Washing Process of Sand Blasting :</u>

- **Aluminum Oxide (Al_2O_3)** is used, which is **very fine, hard, and sharp**.
- **Al_2O_3 is sprayed through a gun** using **dry air** to create the blasting effect.
- The **gun** has an **on/off switch** to **control the sand flow**.
- **Garments are placed on a bed inside a closed chamber** during blasting.

- **Fading happens** because of the **frictional impact of Al_2O_3** on the fabric.
- **Blasting angle** is kept at **10° to 20°** for proper fading.
- **Operators must wear safety gear**:
    - Protective **mask**, **gloves**, **helmet**, **earflap**, and **uniform**.
- After mechanical fading, chemicals treatment are required.

Precaution of Sand Blasting :

Proper protection is needed for the operator so that any threat can be avoided :

• Protection eyeglass,

• Using mask,

• Earplug,

• Should use a helmet, Cap.

• Using apron

Demerits/Disadvantages of Sand Blasting :

- ✕ **Not Environment-Friendly:**
  Sand blasting creates a lot of dust that pollutes the air and is harmful to the environment.
- ✕ **Health Hazardous – Harmful for Workers:**
  The dust from sand blasting is very bad for workers' health. It can cause breathing problems and dangerous diseases like **silicosis** (lung disease) and even **cancer**.
- ✕ **Chance of Fabric Damage:**
  If the sand is blasted too strongly or at the wrong angle, it can **damage the fabric**, making it weak or torn.
- ✕ **Now Restricted in Many Factories:**
  Many buyers and factories **ban sand blasting** because of serious health risks. Most buyers **do not allow sand blasting** anymore.
- ✕ **Risky Handling Process:**
  The operator must use **protective equipment** like **masks, gloves, helmet, gown**, because the process is risky and creates strong backward air force.
- ✕ **Special Equipment Needed:**
  Sands pass into the machine by regulating system. The machine uses a **blasting regulator** (to control sand amount) and **hand regulator** (to control air pressure), which makes it more complicated to use.
- ✕ **Fabric Angle Control Required:**
  If the **blasting angle is too high**, it creates **strong fading but also high risk of damage**. Workers need skill to handle it properly.

- ✕ **Fading Only on Selected Areas:**
  The process is done on **pre-marked areas only**, and needs careful attention.
- ✕ **Low Production Rate:**
  Only **25 to 45 garments per hour** can be processed depending on fading amount, which is **slow compared to other methods**.

Whiskering -

Whiskers are one of the most popular designs in making jeans look old or used.The idea comes from the **natural faded lines** that appear on the **hip and thigh area** after wearing jeans for a long time.These lines or patterns depend on things like the **fabric quality**, **body shape**, and **sitting position** of the person.Whiskers make the jeans look **fashionable and stylish**, like they've been used for a long time.Whiskers are also called **"Cat's Whiskers"** because the lines look like the whiskers of a cat.

Whisker Making Process :

- ✓ **Step 1:** First, the **buyer's design** is drawn on a **tracing paper**.
- ✓ **Step 2:** This design is **transferred from the tracing paper onto plywood**.
- ✓ **Step 3: Foam rubber** is cut and **pasted on the plywood** following the design using **solution gum** (glue).
- ✓ **Step 4:** The **top side of the foam rubber is shaped (sloped or round)** by rubbing or grinding.
- ✓ **Step 5:** This **pattern board** is put **inside the front part of the jeans**.
- ✓ **Step 6: Rubbing with sandpaper (320-500 grit)** is done carefully on the jeans over the pattern to create the **whisker lines**.
- ✓ **Step 7:** Sometimes it takes **3 or more tries** to get the whisker design to **match the buyer's exact style**.

Result: After rubbing, the faded lines that look like natural folds are called **whiskers**.

Hand Scraping -

☞ **What is Hand Scraping?**
Hand scraping means **rubbing the surface of jeans with sandpaper (emery paper)** to make some parts look **faded or used**.

☞ **Why it is done?**

- To **visible a light faded area** where the jeans usually get worn out, like the **front thighs or knees**.
- Sometimes it also helps **show the inside pocket shape** from the outside.
- It is also used as a **safe alternative to sand blasting** (which is harmful for health).

☞ **How it is done?**

Workers **rub jeans by hand** with **emery paper** on specific areas. This creates a **natural faded look**, like **old or used jeans**.

Destroy -

It looks like holes and worn out white yarns which make garment unique

and fashionable

Destroy Process

➞ **Destroy by Machine:**

- **Step 1:** First, **mark the area** on the jeans where you want to **make destruction (damaged effect)**.
- **Step 2:** Use a **grinding machine** and **slightly attach** the machine on that area.
- **Step 3:** The machine slowly **removes the cellulose (warp yarn)** from the fabric.
- **Step 4:** After wet wash, this area will get the desired effect .

➞ **Destroy by Needle and Knife (Manual Method):**

- **Step 1:** Workers use a **knife or needle by hand** to **cut the warp yarn** or make **small holes**.
- **Step 2:** Normally, **3 to 5 small holes** are enough on a jeans.
- **Step 3:** The **weft yarn (width threads)** is kept **unbroken** to show the **white thread lines**.

- **Step 4:** The **knife is run along the warp direction** around **10 or more times**.
- **Step 5:** The area around the destruction will also look **old and worn out** after washing.

✓ **Result:** After washing, the jeans will show the desired effect .

Potassium Permanganate (P.P) Sponging/Brushing) :

- P.P spray is done on denim clothes to make certain small areas look lighter or whiter than the rest of the blue fabric.
- This is done by dipping sponges into the P.P solution and rubbing them on the parts that need to be faded.
- After rubbing, the garment is treated with neutralizing chemicals in a wet wash to stop the chemical reaction.
- This hand rubbing can be done after the clothes are stiff (rigid) or during the washing process.
- Doing it after enzyme or bleach washing gives a more natural and soft white effect than doing it on stiff clothes.
- Sometimes, extra chemicals (additives) are added to get the exact look or intensity wanted.
- Usually, painters' brushes are used, and sometimes the brush hairs are cut in different shapes to make special designs.
- Towels, sponges, and straw bunches can also be used to create different effects.
- The best and most natural-looking effects are made using towels dipped lightly in the solution and gently rubbed over the garment.
- This creates random, beautiful faded patterns that look great on dark denim.
- This process is difficult and needs skilled workers to do it well.
- Right after the rubbing, the clothes must be neutralized immediately to stop further chemical action.

Procedure of P.P Sponging/ Brushing:

 Prepare Potassium Permanganate solution.(P.P 20%,water 80%).

 Take the solution in a tray or bowl.

 Take a rubber board.

 Then take a jeans trouser & place the rubber board inside the leg panel of the trouser.

 Now place the trouser with rubber board on a flat table & sponged or rubbed P.P on apparel surface.

 Then color will faded.

 After this process wet process essential (detergent/bleach/enzyme wash).

P.P (Potassium Permanganate) Spray Process –

- P.P spray is done inside special spray rooms called **spray booths.**
- **Rubber dummies** are used to hold the garments, and **air is pumped** inside the dummy so the clothes stay stretched and tight.
- Different dummies are used for different clothes like kids' clothes, men’s jeans, jackets, shirts, etc.
- When workers spray Potassium Permanganate (P.P) on jeans, it creates **chemical air (bad smell).**
- The chemical air is sucked out by an **air exhaust system**.
- Then the chemical air is sent to a **water shower**, where the water cleans the air.
- Clean air is **released outside,** so it doesn’t pollute the environment.
- The water inside the shower gets dirty because it catches the chemicals.
- This dirty water is cleaned by a **neutralizer chemical** before it goes to the **drain.**
- But if low-strength P.P solution is used, the water doesn’t need **special treatment.**

- This P.P mixed water can also be reused to clean other water areas **because it helps to kill germs.**

Laser fading -

Laser fading (Marking) is a popular dry process for denim now a days It is used a lot now because it **replaces old methods** like sandblasting, which can be **dangerous for health**. Apart from this, Laser systems are used in fashion design, pleating, cutting and modification of fabric surface to impart some special finish. Laser fading works with better precision but also have some drawbacks.

Laser Machine Main Parts :

✓ **Laser Beam System:**
☞ This part makes and controls the laser light (beam) to create designs.

✓ **Work Bed:**
☞ A flat table where the jeans or fabric are placed for marking or design.

✓ **Fume Extractor:**
☞ A system that removes bad smoke or fumes created during laser work, so the air stays clean.

✓ **Outer Cover (Housing):**
☞ A protective box that keeps the laser machine safe and covers all the parts inside.

Mechanism of Fading by Laser :

✓ ☞ **Step 1:** The laser focuses on a small area of the fabric.

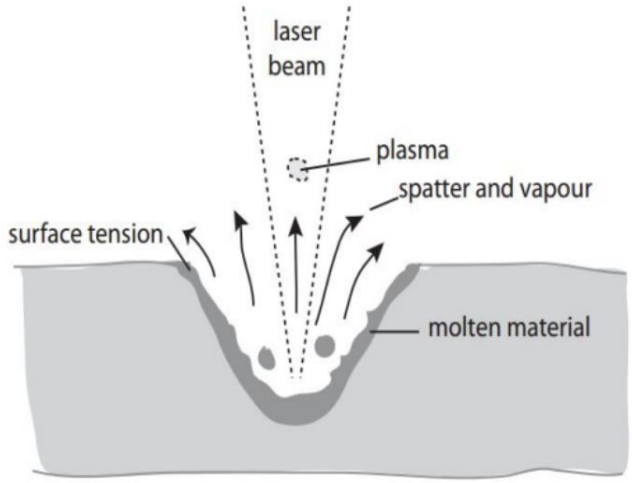
✓ ☞ **Step 2:** The laser energy heats that small area very quickly.

✓ ☞ **Step 3:** Because of the high heat, the fabric melts in that spot (solid turns into liquid).

✓ ☞ **Step 4:** Some of the melted part moves around because of liquid surface tension.

✓ ☞ **Step 5:** The rest of the melted fabric gets even hotter, turns into gas (boils), and fades the fabric color.

This fast heating and vaporizing process removes color in certain areas, making fading designs on denim.



✓ **Laser Fading Procedure on Denim :**

1. **Design Creation:**
   A design pattern is created using graphic editing software. Sometimes, a photo is captured and converted into a design.
2. **File Conversion:**
   The design is converted into grayscale format and saved as a **.bmp (bitmap image format)** file, which is suitable for the laser machine.
3. **File Input:**
   The **.bmp** design file is uploaded into the laser system for processing.

4. **Laser Setting Adjustment:**
   Important laser settings such as **pixel time** (duration of laser focus on a point) and **resolution** (clarity of design) are set according to the desired effect.
5. **Garment Placement:**
   The garment is placed properly on a **honeycomb cutting table**. The target area for laser engraving is marked.
6. **Laser Engraving:**
   The laser machine performs engraving on the selected area, producing a faded effect on the fabric by removing the dye.

   Laser fading provides accurate and fashionable faded effects on denim without using harmful chemicals.

Advantages of Laser Fading :

• Any design at any place

• Faster process

• High accuracy

• Suitable for wide range of material

• Comparatively less strength loss

• Low hairiness

• Environment friendly

• Zero water waste

• Requires no chemical

• Less manpower required

Disadvantages of Laser Fading :

- **High Initial Cost:**
  The machine setup cost for laser fading is very expensive.
- **Need for Skilled Operator:**
  Trained and skilled operators are required to handle the machine, which can be a problem in some factories.
- **Difficult to Produce Natural Look:**
  Sometimes it is hard to achieve natural fading effects like hand processes.

- **Health Hazard:**
  The laser beam and fumes created during operation can be harmful if proper safety systems are not used.
- **Complex Maintenance:**
  Regular maintenance and servicing of laser machines can be complicated and costly.